

Apoorva Narula

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[Google Scholar: https://scholar.google.com/citations?user=cupIUlIAAAAJ](https://scholar.google.com/citations?user=cupIUlIAAAAJ)

[Personal Webpage: https://apoorvagor.github.io](https://apoorvagor.github.io)

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Aug 2024 – May 2028 (Expected)

Ph.D., Industrial Engineering (GPA: 4.0/4.0)

Research interests: Mixed-Integer Programming, Change-point detection, Scalable Optimization.

Indian Institute of Technology (IIT) Delhi, India

Jul 2019 – May 2024

B.Tech., Production & Industrial Engineering (GPA: 9.29/10)

Institute Silver Medalist – Highest CGPA in B.Tech. cohort

M.Tech., Industrial Engineering (GPA: 9.81/10)

Highest CGPA in M.Tech. cohort - IIIE Cash Award

PhD Research Statement

My PhD research focuses on developing scalable optimization methods for change-point detection in time-series data. In many datasets and applications (such as manufacturing quality control, sensor-based monitoring, and climatology), changes do not always appear as abrupt jumps; instead, signals may begin to increase or decrease at a different rate, resulting in more subtle slope changes that can be difficult to detect. We develop optimization-based methods for detecting such changes through continuous piecewise-linear fitting. For the offline multiple slope-change problem, where a univariate time series undergoes multiple changes in trend, we develop strengthened mixed-integer programming formulations for optimally partitioning the time domain into contiguous linear segments while maintaining continuity at the breakpoints. We also study slope-change detection in high-dimensional time series, where multiple signals may share a single common change point. For this setting, we establish polynomial-time solvability under several error norms and develop scalable domain-reduction techniques for solving sparse and non-sparse variants in very high dimensions. Overall, my research combines discrete optimization, statistical learning, algorithmic techniques, and parallel computation to develop change-point detection methods with theoretical guarantees and strong computational performance.

PAPERS & PRE-PRINTS

- [1] Apoorva Narula, Santanu S. Dey, and Yao Xie. *Mixed-Integer Programming for Change-point Detection*. 2026. arXiv: 2602.11947 [math.OC]. URL: <https://arxiv.org/abs/2602.11947>.

SKILLS

Operations Research: Mixed-integer programming, Change-point detection, Discrete event simulation.

Data Science: Hands-on experience in training deep learning models and time series analysis.

Programming: Python, Gurobi, C++, MPI

WORK EXPERIENCE

Georgia Institute of Technology, Atlanta, GA

Fall 2024 – Fall 2026

Graduate Teaching Assistant

- Teaching Assistant for Deterministic Optimization, Foundations of Modern Data Science, and Regression, with class sizes varying from 20 to 150 students.

Argonne National Laboratory, Lemont, IL

Summer 2025

Research Aide (PhD)

- Developed interpretable predictive models linking historical power outages with meteorological and environmental factors to inform grid resilience planning.

Indian Institute of Technology (IIT) Delhi, New Delhi, India

2022 – 2024

Teaching Assistant

- Assisted with undergraduate optimization courses through weekly recitations and assignment grading for approximately 150 students.

Tata Institute of Fundamental Research (TIFR), Mumbai, India

Summers 2021, 2023

Research Assistant

- Summer 2023: Developed deep learning models using spatial and temporal weather patterns for rainfall prediction, achieving 20% lower error than ECMWF-HRES for 1–3 day forecasts across India.
- Summer 2021: Modeled district-level oxygen demand–supply gaps during the COVID-19 pandemic.

PAST RESEARCH EXPERIENCE

- Minimal-Contact Routing of Randomly Arriving Agents** B.Tech. Thesis, Prof. Varun Ramamohan, IIT Delhi
- Developed robust mixed-integer optimization models for congestion-aware routing and throughput maximization in dynamic logistics networks, with applications to AGV routing.
- Approximation Algorithms for Optimized Scheduling** Independent Research, Prof. Naveen Garg, IIT Delhi
- Studied approximation algorithms for Open Shop Scheduling and proved NP-hardness of the Data Migration problem on trees with arbitrary edge lengths, with applications to resource allocation and scheduling.
- Sentiment-Driven Stock Prediction** Independent Research, Prof. Kiran Seth, IIT Delhi
- Developed deep learning sequence models using word embeddings and Twitter sentiment to predict stock-market trends from textual information.

TALKS & POSTERS

- 2026 NeurIPS ML×OR Workshop**, Atlanta, GA December 2026
Poster: *Very High-Dimensional Slope Change Detection with Guarantees*
- 2026 INFORMS Annual Meeting**, San Francisco, CA November 2026
Invited Talk: *Mixed-Integer Programming for Slope Change Detection*
- 2026 ML@GT Student Conference**, Atlanta, GA October 2026
Poster: *Change-Point Detection via Piecewise Linear Fitting Using MIP*
- 2026 Georgia Statistics Day**, Atlanta, GA October 2026
Poster: *Mixed-Integer Programming for Change-Point Detection*
- 2026 MIP Society Workshop**, Stamford, CT May 2026
Poster: *Mixed-Integer Programming for Change-Point Detection*
Best Poster Competition Finalist; Travel Grant Award
- 2026 INFORMS Optimization Society Conference**, Atlanta, GA March 2026
Invited Talk: *Mixed-Integer Programming for Changepoint Detection*
- 2025 NeurIPS ML×OR Workshop**, San Diego, CA December 2025
Poster: *Mixed-Integer Programming for Change-Point Detection*
- 2025 NeurIPS Workshop on Tackling Climate Change with Machine Learning**, San Diego, CA December 2025
Poster: *Deep Learning for Short-Range Monsoon Rainfall Forecast Using Ground-Truth Rainfall Data*
- 2025 INFORMS Annual Meeting**, Atlanta, GA October 2025
Invited Talk: *Spatio-Temporal Statistical Analysis of Power Grid Resilience to Extreme Weather Conditions*

AWARDS & SCHOLARSHIPS

Stewart Fellowship (Georgia Institute of Technology - Fall 2024 and Spring 2025); DAAD WISE Scholar (Kiel University); NTSE scholar (National Talent Search Examination, among selected 1000 scholarship winners out of approximately 3M aspirants); KVPY rank 278 (National Science Fellowship, selected out of approx 1M applicants); JEE Main (Joint Entrance Examination Main, 1.2M applicants): 99.87 percentile; qualified JEE Advanced (next stage of JEE Main), securing admission to IIT Delhi; CBSE National Certificate of Merit (100% Score in Computer Science Exam; 96% aggregate across all subjects including Chemistry, Physics, Mathematics and English).

EXTRA CURRICULAR ACTIVITIES

- Georgia Institute of Technology**, Atlanta, GA 2025 – Present
Bee Well Student Ambassador
- Supporting graduate student well-being by organizing student community events.
- Rohini Ghadhiok Foundation**, New Delhi, India 2017 – 2018
Volunteer Teacher
- Taught English and Mathematics to underprivileged children; designed engaging practice modules.